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TEC–601

B. TECH. (ECE) (SIXTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

MICROWAVE ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the process of setting up the microwave bench setup for maximum deflection with the help of the diagram of microwave bench setup. (CO1)

OR

(b) An air filled rectangular waveguide of inside dimension 7.5×4.5 cm operates in the dominant TE_{10} mode. Find the cut-off frequency. (CO1)

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2. (a) Explain the process of the measurement of frequency, guide wavelength and VSWR using microwave bench setup. (CO2)

OR

(b) Derive the expressions of electric field and magnetic field in TE mode propagation in rectangular waveguide. (CO2)

3. (a) Explain the following terms of rectangular waveguide briefly : (CO2)

(i) Cutt-off frequency

(ii) Dominant mode

(iii) Phase velocity and group velocity

(iv) VSWR

(v) Reflection coefficient.

OR

(b) Prove that three port network cannot satisfy all three conditions of S-matrix at a time simultaneously. (CO2)

4. (a) Explain Magic Tee with the help of its scattering matrix. (CO1)

OR

(b) A TE_{11} mode is propagating through a circular waveguide. The radius of the guide is 11 cm and the guide contains an air dielectric. Find cut-off frequency. (CO1)

5. (a) Prove that for matrix of directional coupler $a^2 + b^2 + 1$, where a and b are real and positive ? (CO1)

OR

(b) Derive the equations for two wire parallel transmission line and its solutions. (CO1)